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IPv4 Packet Header

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The IPv4 header is a 20-60 byte structure that contains information fields about the packet at the beginning of every IPv4 packet. It acts like a shipping label, telling routers where the packet is destined and how to handle it.



IPv4, see our [Introduction to IP Version 4](#) lesson. Let's start with this picture:

Version	IHL	Type of Service	Total Length	
Identification			Flags	Fragment Offset
Time to Live	Protocol		Header Checksum	
Source Address				
Destination Address				
IP Options (if any)				
Data				

This picture contains all the fields you find in an IPv4 packet header. The blue items are the **fixed header** with the mandatory fields. Every IP packet requires these fields. The header grows when you add IP options. Most packets won't need any IP options, so the default header is fine.

-
- IPv4 packets are stateless. Each packet is independent and  unaware of other packets or connections. Connection tracking occurs at [higher layers](#), such as TCP.
-

1. Header Fields

Let's dive into each header field.

1.1. Version

The first field tells us which IP version we are using; only IPv4 uses this header, so you will always find the decimal value 4 here. The IPv6



The minimum length of an IP header is 20 bytes, so with 32-bit increments, you would see a value of 5 here. The maximum value we can create with 4 bits is 15, so with 32-bit increments, that would be a header length of 60 bytes. This is the absolute limit of the header length. This field is also called the **Internet Header Length (IHL)**.

1.3. Type of Service (ToS)

This field is used for [Quality of Service \(QoS\)](#). There are 8 bits that we can use to give the packet a certain treatment in our network. This is useful because not all packets are equal. Some of your packets might be for Voice over IP (VoIP) traffic, and other traffic could be less important, such as file downloads. In the original IPv4 specification (RFC 791) from 1981, this field is named the Type of Service (ToS) field. Later, in 1998, the ToS field was deprecated and redefined, and split into two in RFC 2474:

- 6 bits for Differentiated Services Code Point (DSCP)
- 2 bits for Explicit Congestion Notification (ECN)

DSCP is used to tell what priority the packet has, as explained in my [IP precedence and DSCP lesson](#). The ECN field enables congestion notification, which allows senders to slow down before packet loss. The receiver can set the ECN bit to tell the sender to slow down.

1.4. Total Length

This 16-bit field indicates the entire size of the IP packet (header and data) in bytes. The minimum size is 20 bytes (if you have no data), and the maximum size is 65535 bytes, that's the highest value you can create with 16 bits.

1.5. Identification



1.6. IP Flags

These 3 bits are used for fragmentation:

- The first bit is reserved and must always be set to 0.
- The second bit, called the **DF (Don't Fragment) bit**, indicates that this packet should not be fragmented.
- The third bit, called the **MF (More Fragments) bit**, is set on all fragmented packets except the last one.

 The DF bit is also used for Path MTU Discovery (PMTUD). If a packet is too large for a link and the DF bit is set, the router drops the packet and sends an ICMP "Fragmentation Needed" message back to the sender.

1.7. Fragment Offset

This 13-bit field specifies the position of the fragment in the original fragmented IP packet. It is measured in 8-byte units. For example, if you see a value of 185 in this field, the data actually starts at byte 1480. $185 * 8 = 1480$. With 13 bits, the highest value we can set is 8191.

$8191 * 8 = 65528$ bytes. This makes sense because the maximum IP packet size is 65535 bytes. A fragment offset of 65528 bytes leaves just enough room ($65535 - 65528 = 7$ bytes) for a final fragment payload of up to 7 bytes.

1.8. Time to Live (TTL)

Every time an IP packet passes through a router, the time-to-live field is decremented by 1. Once it reaches 0, the router will drop the packet and send an ICMP time-exceeded message to the sender. The time-to-live (TTL) field has 8 bits and prevents packets from looping forever (if



1.9. Protocol

This 8-bit field tells us which protocol is encapsulated in the IP packet. Here are some examples:

- ICMP: 1
- TCP: 6
- UDP: 17

1.10. Header Checksum

This 16-bit field stores a checksum of the header. The receiver can use the checksum to detect errors in the header. This checksum **does not validate the payload data**. Upper-layer protocols, such as [TCP](#) or [UDP](#), can perform data integrity. When a router forwards an IP packet, the TTL decrements by one, and the router **must** recalculate the header checksum. This recalculation happens at **every router hop**. This way, header integrity is ensured throughout the packet's journey.

1.11. Source Address

Here you will find the sender's 32-bit source IP address.

1.12. Destination Address

Here you will find the receiver's 32-bit destination IP address. Routers use this to make [forwarding decisions](#).

1.13. IP Option

This field is rarely used, optional, and has a variable length depending on the options used. When you use this field, the value in the header length field will increase. Here are some examples:



Many routers or firewalls drop IP packets with IP options set for security and performance reasons. IP options can expose network topology, which is why they are often blocked for security reasons. IP packets with IP options are processed by the router's CPU, which is why they are often blocked due to performance reasons.

2. Wireshark Capture

When we look at the IP header, the fields are transmitted in a specific sequence. We call this **Network Byte Order**. It uses **Big-Endian** format, which means the most significant byte is sent first. This is important because different computer processors store data differently in memory. By enforcing a standard order, we ensure that every router, switch, and server interprets the IP header values exactly the same way.

Here's what an actual packet looks like:

```
Frame 3: Packet, 114 bytes on wire (912 bits), 114 bytes
captured (912 bits) on interface eth1, id 0
Ethernet II, Src: aa:bb:cc:00:01:10 (aa:bb:cc:00:01:10),
Dst: aa:bb:cc:00:02:10 (aa:bb:cc:00:02:10)
Internet Protocol Version 4, Src: 192.168.12.1, Dst:
192.168.12.2
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN:
Not-ECT)
        0000 00.. = Differentiated Services Codepoint:
Default (0)
        .... ..00 = Explicit Congestion Notification: Not
ECN-Capable Transport (0)
    Total Length: 100
```



```
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 255
Protocol: ICMP (1)
Header Checksum: 0x2245 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.12.1
Destination Address: 192.168.12.2
[Stream index: 0]
Internet Control Message Protocol
```

Let me explain what we see above:

- **Version: 4** – Indicates this is an IPv4 packet (as opposed to IPv6)
- **Differentiated Services Field: 0x00** – Used for Quality of Service (QoS); value 0x00 means default/best-effort delivery with no special priority.
- **Total Length: 100** – The entire IP packet size is 100 bytes, including header and data.
- **Flags: 0x0** – Control bits for fragmentation; all set to 0 means the packet is not fragmented, and fragmentation is allowed.
- **Fragment Offset: 0** – Indicates the position of this fragment in the original packet; 0 means this is either the first fragment or the packet is not fragmented.
- **Time to Live: 255** – Maximum number of hops (routers) the packet can pass through before being discarded; starts at 255.
- **Protocol: ICMP (1)** – Identifies the protocol used in the data portion of the packet; value 1 indicates **ICMP (Internet Control Message Protocol)**, commonly used for ping.
- **Header Checksum: 0x2245** – Error-checking value for the IP header to detect corruption during transmission.
- **Source Address: 192.168.12.1** – The IP address of the device sending this packet.
- **Destination Address: 192.168.12.2** – The IP address of the device that should receive this packet.



3. FAQ

- ▶ **Can the TTL field be used to determine the operating system of the sender?**
- ▶ **Does the Source or Destination IP address ever change while the packet is traveling?**
- ▶ **If the maximum packet size is 65,535 bytes, why are most packets on the Internet much smaller (usually 1500 bytes)?**
- ▶ **How does the router handle the “Options” field if the header length is variable?**

4. Additional Resources

If you want to dive deeper into the specifications, you can take a look at these RFCs:

- [RFC 791](#): The original Internet Protocol specification that defines the IPv4 header structure.
- [RFC 2474](#): Redefined the Type of Service field into DSCP for modern QoS.
- [RFC 3168](#): Added Explicit Congestion Notification (ECN) to the header.
- [RFC 6864](#): Updated guidance on the Identification field (use only for fragmented packets).
- [RFC 1340](#): IANA-assigned protocol numbers.

Want to look at some more IP packets?

Take a look at the fragment offsets here:

Packet Capture: Fragmented Packets

[Packet Capture: Traceroute on Linux](#)[Packet Capture: Traceroute on Windows](#)

5. Conclusion

I hope this lesson has helped you understand the different fields in the IPv4 packet header. If you want to put the theory into practice, take a look at a couple of different packet captures with Wireshark and inspect the IP packet yourself. If you have any questions, feel free to leave a comment.

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Introduction to IP (Internet
Protocol) Version 4

Next Lesson »
Basics of Binary Numbers

Forum Replies



ashoklukose

Hello Rene,
Could you please help on below queries

How network layer decides whether a packet to be fragmented or not?
Is it possible for a admin/application to enforce a fragmentation decision on IP packet?
Is there any relation between total length field and mtu?



I wrote a post on this, I think it will answer your questions:

<https://networklessons.com/ip-routing/pppoe-mtu-troubleshooting-cisco-ios>

Rene



ashoklukose

Thanks a lot Rene.. It was well explained there...



talk2seeni

Hi Rene,

Why Header maximum length limited to 24 bytes ?

As per the theory, For 5 bit header field , maximum value is 15 so header length will be $4 * 15 = 60$ bytes.

Please correct me if it is limited to 24 bytes in production.

Regards,

SV



ReneMolenaar

Hi SV,

Good question, some sources (for example Routing TCP/IP Volume 1) state that the maximum header length is 24 bytes so that 6 would be the

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Rene

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